CO2 AT2 - Scenario Based Questions

DSA 0201 – Computer Vision with Open cv

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1. Crack Detection on Concrete Surfaces Using Drone Images

A. Image Preprocessing Pipeline

To accurately detect thin cracks on concrete surfaces, the drone images should undergo the following preprocessing steps:

Step 1: Illumination Correction

Method: Histogram Equalization or Adaptive Histogram Equalization (CLAHE)

- Drone images are captured under different lighting conditions.
- Shadows and bright sunlight create uneven illumination.
- Illumination correction produces a more uniform image, making cracks easier to detect.

Step 2: Noise Reduction

Method: Gaussian Filter

- Drone camera vibration introduces Gaussian noise.
- Noise may create false edges during edge detection.
- Gaussian filtering smooths the image while preserving important crack structures.

Step 3: Contrast Enhancement

Method: Contrast Limited Adaptive Histogram Equalization (CLAHE)

- Thin cracks usually have very low contrast.
- CLAHE enhances local contrast without over-amplifying noise.
- Fine cracks become more visible against the concrete background.

Step 4: Image Normalization

Method: Min-Max Normalization (pixel values scaled between 0 and 1 or 0 and 255)

- Images captured under different lighting conditions have different intensity ranges.
- Normalization provides a consistent intensity scale.
- It improves the performance of edge detection and machine learning algorithms.

B. Edge Detection Technique

Selected Technique: Canny Edge Detection

Reasons for Selection

- Detects very thin and weak edges accurately.
- Uses Gaussian smoothing to reduce noise before edge detection.
- Applies non-maximum suppression to produce thin edges.
- Uses double thresholding and edge tracking to reduce false detections.
- Produces continuous and accurate crack boundaries.

Comparison with Sobel Edge Detection

Canny Edge Detection	**Sobel Edge Detection**
Better noise removal	Sensitive to noise
Detects thin cracks accurately	May miss fine cracks
Produces thin edges	Produces thicker edges
Double threshold reduces false edges	Generates more false edges
Higher detection accuracy	Lower accuracy for small cracks

Why Canny is More Suitable

Thin cracks are narrow and have weak intensity differences. Canny edge detection is more suitable because it detects weak edges while suppressing noise, resulting in accurate and continuous crack detection.

C. Post-Processing Technique

Technique: Morphological Closing

Operation:
Closing = Dilation followed by Erosion

Purpose

- Connects broken crack segments.
- Fills small gaps between crack edges.
- Produces continuous crack structures.
- Improves the overall detection result.

Why It Is Needed

After edge detection, cracks may appear disconnected due to shadows, noise, or uneven illumination. Morphological Closing joins nearby crack segments and creates a complete crack pattern, making further analysis more reliable.

2. Human Motion Tracking in a Security Camera Video

A. Lucas-Kanade Optical Flow Method

The Lucas-Kanade optical flow method estimates the motion of objects by tracking the movement of pixels between two consecutive video frames.

Working:

1. Capture two consecutive video frames.
2. Detect feature points such as corners or edges on the moving person.
3. Assume that the brightness of the same point remains constant between frames.
4. Calculate the displacement (motion) of each feature point within a small neighbourhood.
5. Generate optical flow vectors that indicate the direction and speed of movement.
6. Use these vectors to continuously track the person's movement across frames.

Why it is Suitable

- Works well for tracking small movements between consecutive frames.
- Efficient and fast for real-time security surveillance.
- Provides accurate motion estimation for moving people.

B. Aperture Effect

The aperture effect occurs when only a small part of a moving object is visible, making it difficult to determine its actual direction of motion.

Effect on Motion Estimation

- Plain walls, smooth floors, and clothing surfaces have very little texture.
- These uniform regions do not provide enough feature points for optical flow calculation.
- The algorithm cannot accurately estimate motion direction.
- As a result, motion vectors may be incorrect, incomplete, or missing.
- This reduces the tracking accuracy of the security system.

Example

If a person is wearing a plain white shirt, the algorithm may struggle to estimate the motion on the shirt because it lacks distinct features.

C. Sudden Change in Speed or Direction

Effect on Optical Flow Vectors

When a person suddenly changes speed or direction:

- Optical flow vectors change rapidly in both magnitude and direction.
- The vectors become irregular.
- Some vectors may be incorrect or lost.

- This can cause temporary tracking errors or loss of the target.

Method to Improve Tracking Accuracy

Method: Kalman Filter

Why Kalman Filter?

- Predicts the person's next position using previous motion information.
- Handles sudden changes in movement more effectively.
- Reduces the effect of noisy or missing optical flow vectors.
- Provides smoother and more accurate tracking.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

A. Optical Flow for Vehicle Motion Estimation

Optical flow estimates the movement of vehicles by calculating the motion of pixels between two consecutive video frames.

Working:

1. Capture two consecutive CCTV video frames.
2. Detect feature points such as vehicle corners, edges, or textures.
3. Compare the position of these feature points in both frames.
4. Calculate the displacement of each point.
5. Generate optical flow vectors showing the direction and speed of each vehicle.
6. Use these vectors to continuously track vehicles throughout the video.

Why it is Suitable

- Estimates both the speed and direction of moving vehicles.
- Works well for real-time traffic monitoring.
- Helps detect vehicle movement, stopping, and turning at road junctions.

B. Challenges Affecting Motion Estimation

1. Occlusion (Vehicles Overlap)

- One vehicle partially or completely blocks another vehicle.
- Some feature points become hidden.
- Optical flow vectors may disappear or become incorrect.
- This can cause the tracking system to lose the vehicle or confuse it with another vehicle.

2. Shadows and Reflections

- Shadows and reflections create false motion in the video.
- The optical flow algorithm may detect shadow movement instead of actual vehicle movement.
- This produces incorrect motion vectors and reduces tracking accuracy.

C. Improvement Method

Method: Kalman Filter

Why Kalman Filter?

- Predicts the future position of each vehicle using its previous motion.
- Continues tracking even when a vehicle is temporarily hidden due to occlusion.
- Handles sudden changes in speed or direction more effectively.
- Reduces the effect of noisy optical flow vectors.
- Provides smoother and more reliable vehicle tracking.